

DATA301 Interim Report

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1. Background

Our group worked on the Steam Games Dataset published on Kaggle by Artemiy Ermilov. The dataset contains information about available games found on the Steam library which is one of the largest digital distribution platforms of games for PC. The dataset contains information on more than 90,000 games released between April 1997 and March 2025. The authors provide both a raw dataset and cleaned version removing duplicate entries and playtest versions of games (playtest games are incomplete versions that are given to testers in order to give feedback to the developers before the final game is released). While a May 2024 dataset is also available, we decided to focus our analysis on the March 2025 data for our analysis.

The version of the dataset used in our analysis is the cleaned 2025 version that contains 89,618 observations and 47 variables. It provides information about individual games and includes attributes relating to their release, pricing, genres, developers, publishers, user reviews, supported platforms and more. Each observation represents a game found on Steam, while the columns are different attributes of that game. This dataset is of interest because it provides an opportunity to investigate patterns in the video game industry and explore factors associated with the popularity and success of games. Steam contains a wide variety of games of differing and distinct characteristics. Therefore, analyzing these characteristics will provide an opportunity to discern whether they can be associated with the greater success of a game. The dataset can be used to explore questions like whether some genres are more successful than others, whether the price of a game is associated with its popularity and whether characteristics such as release year or specific developers are related to the number of positive reviews a game receives. These questions allow us to use exploratory data analysis to identify patterns and relationships within the dataset rather than focusing on a single type of game.

1a. Data Types and Structure

The dataset contains a mixture of numerical, categorical, date and text based variables. Numerical variables include measures such as game price and review related statistics. Categorical variables describe characteristics such as the game's genre, developer, publisher and supported platforms. The dataset also contains date information including game release dates which allows games to be compared across different periods. Each row represents one game rather than a repeated measurement over time meaning that the dataset is not a traditional time-series dataset. The release date variables allow trends in the number and characteristics of games to be investigated over time. Some variables can also contain multiple categories for a single game. For example, a game may fall under multiple different genres so this needs to be considered when summarising categorical variables.

1b. Variable Selection and Data Cleaning

Before conducting our analysis, we removed several variables that we thought were not relevant and would not provide useful information. These included `header_image`, `website`, `support_url`, `support_email`, `metacritic_url`, `screenshots`, `movies`, `packages`, `detailed_description` and `about_the_game`. Variables containing URLs and email addresses, we believe won't provide any insightful information pertaining to our analysis as they serve to redirect to outside sources. Variables that represent visual elements like images and videos won't be used as we will focus on using numerical and text data for the analysis. There are three variables that contain descriptions of each game which have different amounts of text. For the purpose of simplicity, the `short_description` will be kept and the other two excluded. The `packages` variable consists of lists of additional content unique to each game, thus, removing the variable reduces unnecessary noise in the dataset. We also excluded `score_rank` due to its extremely high number of missing data. Of the 89,618 games in the dataset, 89,579 games had no score rank recorded which makes up 99% of games in the dataset. This leaves only 39 observations with available values. Including this variable would provide very limited information and could substantially reduce the number of observations available for analyses involving it.

1c. Completeness and Data quality

We assessed the completeness of the dataset by examining missing values and duplicate observations. The dataset contains 89,618 rows and 47 columns and no duplicate rows, we also found that substantial amounts of missing data were present in some variables. For example, the `reviews` variable contains 79,217 missing values, representing 88.39% of observations. The high proportion of missing review data is important when using reviews as an indicator of game popularity as analyses based on this variable may represent only a subset of the games in the dataset.

Approximately 10,401 games have recorded review information, meaning that conclusions involving reviews need to be interpreted with this limitation in mind. Missing data was considered when selecting variables for analysis and interpreting results. Rather than assuming that missing values represented zero, missing observations were treated as unavailable information unless there was a clear reason to interpret them otherwise. As no duplicate rows were identified, duplicate observations did not require removal during the data cleaning process. Other similar data preparation steps were performed when necessary to ensure that variables were in suitable formats for analysis. Because our analysis is solely focused on the Steam Games Dataset, no integration of multiple datasets was required.

2. Ethics, Privacy and Security

When looking into ethical considerations of using any dataset for data analysis, we must always check how the data was collected. According to kaggle, who hosts the Steam games 2025 dataset, the data was collected directly from scraping the Steam store page, the Steam API and public user metrics (reviews, scores etc.), which all fall under publicly available information. Therefore, there are few ethical concerns when it comes to the data collection process. The Steam games dataset also uses plenty of copyrighted material and intellectual properties. This includes game titles, descriptions, screenshots from the games, artwork and reviews. While artwork and screenshots will likely not be used during the analysis, we may utilize material like descriptions and reviews. Thus, we will keep in mind to use these materials strictly for analysis and we will not republish them in any way.

Another ethical concern is inherent bias within the dataset, in particular, when using variables such as reviews and scores in the analysis. In most cases, a large number of people who buy games often don't bother to give a rating and an even smaller minority are vocal enough to write a review. Thus, these metrics are not indicative of the whole population of players who own any particular and in some cases, skew the reality of how people perceive a game. Furthermore, there is also bias linked to the differing popularity of games on Steam. Games with larger playerbases will always have more data than lesser known games. Additionally, those more niche games potentially have dedicated communities which may skew results.

A major ethical concern in this project is the privacy of the Steam users in the dataset. While personal information from Steam users accounts isn't part of any variable within the dataset, we cannot ignore the likelihood of users including personal information in their submitted reviews. Our analysis project does not necessitate the use of such private information. Therefore, when this information does come up, we will do our best to ignore or omit them during our analysis and model building. To keep our project and results secure, we will maintain all the details within group members and our supervisors. Our reports will be done on a Google docs that has access restricted to group members and supervisors. Discussions about the project are held within private group chats and meetings. The codes and resources used will remain within the access restricted to the group.

3. Exploratory Data Analysis

3a. Scoping Data

Our dataset covers games available on Steam that were released from April 1997 to March 2025. The majority of games in the dataset were released after 2016. Figure 1 uses a log scale on the count axis. Without changing the scale, the smaller counts of games released before 2010 look like zero compared to the 2024 peak. The log scale makes the early years visible while also compressing the visual gap between years. Bar height alone should not be used to compare counts. Exact values should be read from the axis.

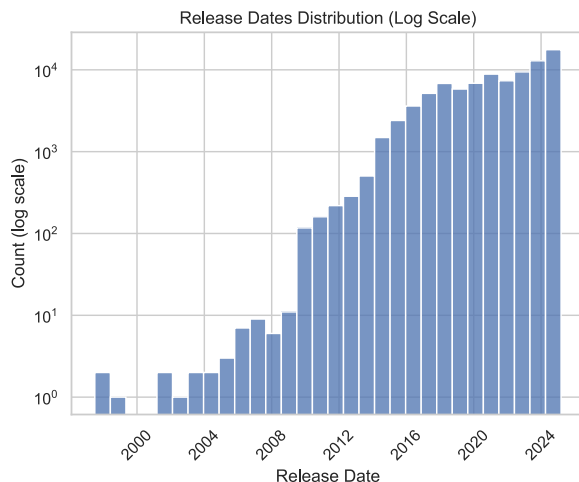


Figure 1: Distribution of game release dates in the dataset, count axis on log

Before further analysis, we filtered out games with fewer than 30 reviews. The dataset sets both `pct_pos_total` and `num_reviews_total` to -1 for games with less than 10 reviews. With cutoff at 10, one review can swing `pct_pos_total` by about 10 points which creates unneeded noise. Figure 2 shows the distribution of review counts, clipped at 200 for readability. Most games sit at the low end of this range, which is why a cutoff matters. A 30 review cutoff keeps a large sample while reducing this noise. This filtering leaves 34,583 games out of 89,618, or 38.6% for remaining analysis.

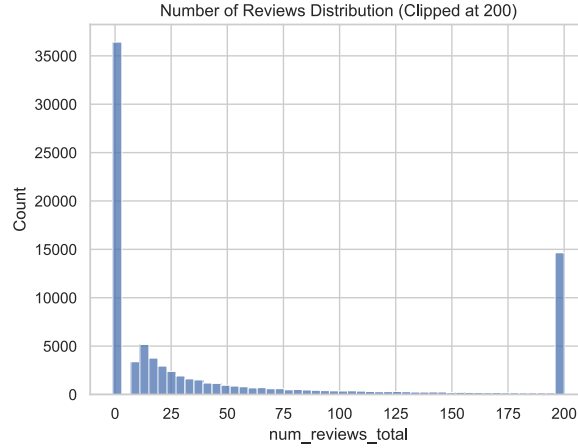


Figure 2: Distribution of total review counts, clipped at 200 reviews for readability

3b. Genre Distribution

The genres column lists every genre associated with a game, and most games have more than one. We use the review-filtered dataset described previously, with some further filtering applied. We dropped 109 games with no genre listed, along with six non-genre labels: “Violent”, “Gore”, “Nudity”, “Sexual Content”, “Free To Play” and “Early Access”. This brings the number of unique genres down from 27 to 21. Figure 3 shows the count of games across the top 15 genres. Indie is by far the most common, with 24,155 games, followed by Adventure and Action at 14,844 and 13,953. The smallest genres here have only a few hundred games each. Since a game with three genres is counted three times, this plot reflects genre labels rather than unique games, and doesn’t show how positive reviews vary within each genre — a minimum size cutoff is applied before that comparison.

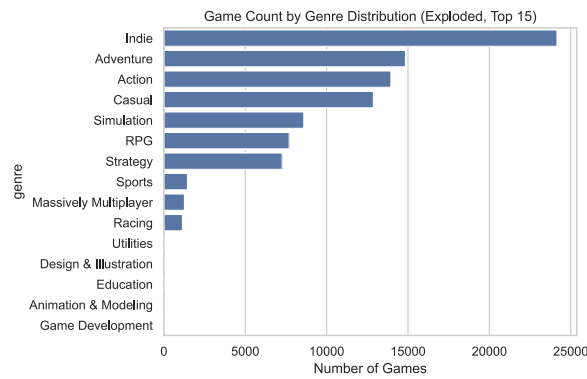


Figure 3: Number of games per genre, top 15 genres (exploded, review-filtered dataset)

3c. Positive Reviews Distribution

The `pct_pos_total` variable is the percentage of positive reviews for a game. We set a minimum size of 100 games per genre before comparing review scores. Without a cutoff, a single game in a small tag could push its median to 100% or 0% and rank it above established genres like Action or RPG. This cutoff removes every software tag and keeps the 10 significant game genres shown in Figure 4. Casual has the highest median at 83%, and Massively Multiplayer has the lowest at 67%. The boxes are notched, showing the confidence interval around each median. Where notches overlap, the difference between two genres' medians is not reliable. Where they don't overlap, as with Casual and Massively Multiplayer, the gap is more trustworthy. The plot does not show how many games sit in each genre. Therefore, we pair it with the `n > 100` filter to avoid assuming equal representation.

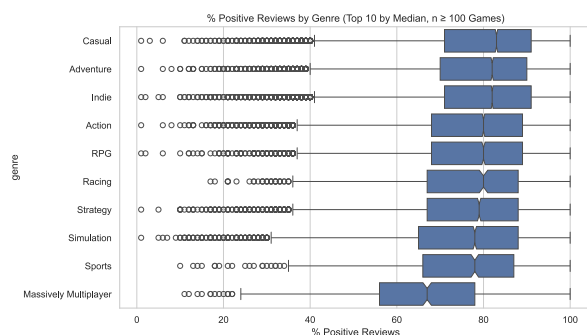


Figure 4: Percentage of positive reviews by genre, top 10 genres with at least 100 games

3d. Confounders (Price and Age by Genre)

The price variable is the base price of a game in US dollars, and `required_age` is the minimum age rating Steam sets for a game. Figure 5 shows the price spread for each genre, with notches marking the confidence interval around each median. Sports and RPG have the highest median price at \$7.99, while Massively Multiplayer sits close to \$0 (this is because most of these games are free to play). Figure 6 shows the share of games with an age restriction in each genre. Action has the highest share at 3.9%, compared to 0.5% for Casual. This pattern lines up with the review score chart in reverse. Genres with a higher price and a higher age-restriction share tend to have a lower median `pct_pos_total`. These two plots show price and age separately by genre. They do not show whether the same games drive both patterns within a genre, only that the genre-level trends move together. This supports the plan to control for both price and age when we study the link between genre and review positivity.

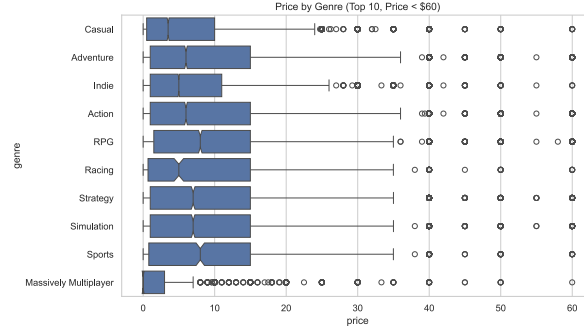


Figure 5: Base price (USD) by genre, top 10 genres, games priced under \$60

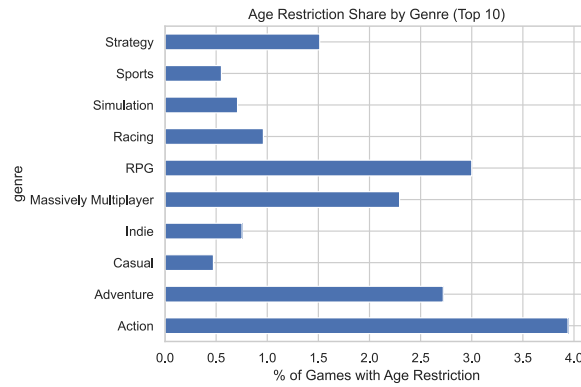


Figure 6: Share of games with an age restriction, by genre, top 10 genres

3e. Distribution of Positive Reviews & Average play time

The `average_playtime_forever` variable is the average number of minutes all players who bought the game have spent in it from when it was added on Steam. The `pct_pos_total` variable measures the percentage of positive reviews. Figure 7 plots these two variables for 34,583 games with 30 or more reviews. `pct_pos_total` ranges from about 13% to 100% in this sample, with most games sitting between 50% and 100% positive. `average_playtime_forever` spans a wide range from 0 minutes up to over a million minutes, so we plot it on a log scale to see the spread clearly. 78.5% of games in our filtered dataset have an average playtime of 0. This is not limited to games near our review cutoff: among these games, review counts range from 30 up to over 2.5 million, with a median of 102. This suggests Steam does not record or report playtime data for every game, so we treat this as a limitation rather than a reason to filter these games out.

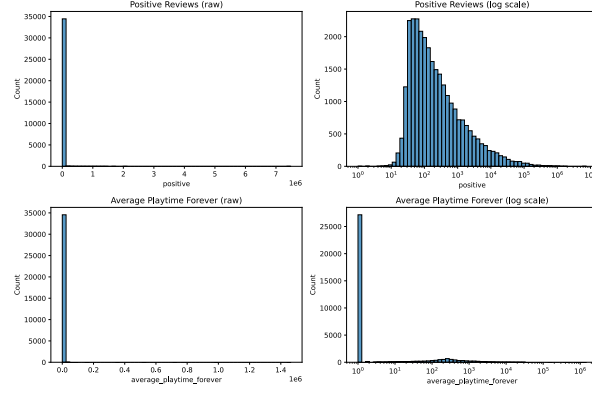


Figure 7: Positive reviews and Average playtime in log scale

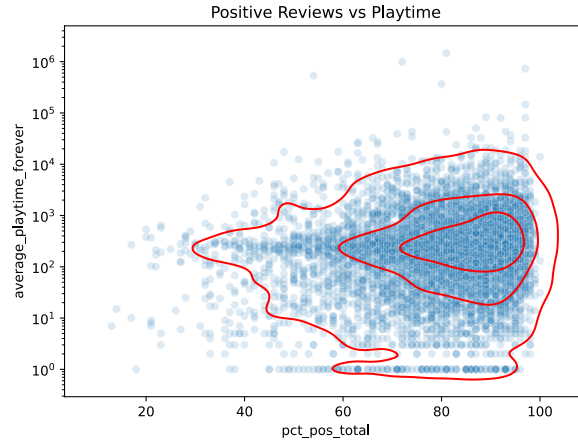


Figure 8: Average playtime (minutes, log scale) versus percentage of positive reviews

3f. Price and Average Playtime

positive is a raw count of positive reviews, unlike the percentage-based `pct_pos_total`. It reflects both how many people reviewed a game and how many rated it positively. The chart below plots price against positive and `average_playtime_forever`. Most games are priced under \$25, with density dropping off beyond \$75 and a few outliers near \$200. Both charts use a log scale, since positive spans single digits to over 10 million and `average_playtime_forever` ranges from near 0 to 10^6 . In both panels, the highest values are concentrated among games priced below \$25, while games above \$75 show a narrower range. The visible band at the bottom of the `average_playtime_forever` panel reflects the large share of games with 0 average playtime, as discussed earlier.

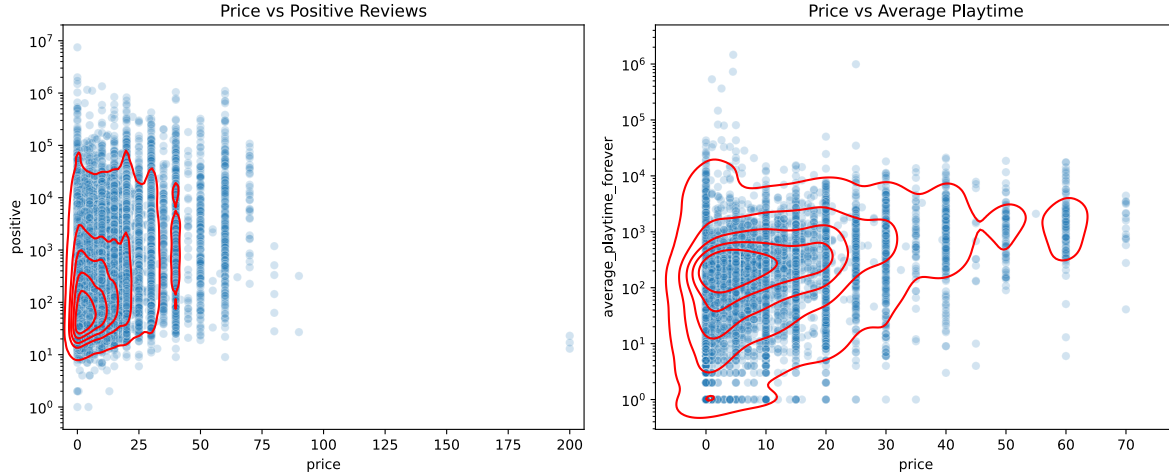


Figure 9: Price vs Positive reviews and Average Playtime

3g. Price and Positive Reviews

To better compare price with positive reviews, we grouped the former into 6 buckets rather than treating price as a continuous variable. We restricted this chart to games priced under \$60 to exclude a small number of extremely high priced outliers. Figure 6 shows the spread of `pct_pos_total` for games in each price bucket. Medians are similar across all six buckets, ranging from 80% to 83%, and every bucket shows a long tail of outlier games well below the main box, most noticeably for games priced at \$40 or more. The boxes overlap heavily across buckets, showing that price alone does not clearly separate games by review positivity.

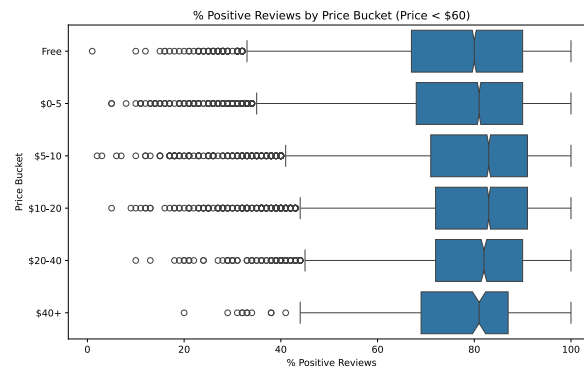


Figure 10: Percentage of positive reviews by price bucket, games priced under \$60

4. Individual Contributions

Section	Contributor
Background	
Data Types and Structure	Joanna
Variable Selection and Data Cleaning	Joanna
Completeness and Data Quality	Joanna
Ethics, Privacy and Security	Derick
Exploratory Data Analysis	
Scoping Data	Rakei
Genre Distribution	Rakei
Positive Reviews Distribution	Rakei
Confounders (Price and Age by Genre)	Rakei
Distribution of Positive Reviews & Average Playtime	Joanna
Price and Success	Joanna
Price and Positive Reviews	Joanna
Additional Tasks	
Proofreading	Derick
Editing	Derick, Joanna
Page Formatting	Derick, Joanna
Appendix A (Project Timeline)	Rakei
Appendix B (Task Allocation)	Rakei
Appendix C (Detailed Data Description)	Derick

5. What's Next?

Joanna's Statement

After performing this EDA, we have found interesting patterns and characteristics of the Steam dataset. For my individual project phase, my original scientific question was “Which game characteristics are strongest predictors of a games success? (Success is measured in positive reviews and average play time)”. We found that this dataset does not provide data for every game that has been played, with a large percentage of the dataset having 0 hours played. I will be changing my success defining variables to better suit the dataset.

Derick's Statement

The insights gained from the EDA have revealed important patterns between Steam games and their unique characteristics that could explain my scientific question of how they can lead to positively received games. To gain better insights on the relationships between variables, I plan to use regression models for columns like price and release date, and classification models for columns like genre and tags. Furthermore, I plan to use text based models to find more complex relationships like specific terms found in the games descriptions or other qualitative factors that can be found in the reviews for the games in the dataset.

Rakei's Statement

This EDA supports my phase 2 question on the relationship between a game's genre and its proportion of positive reviews, after accounting for price and age rating. The confounder checks here show price and age both move with genre in a way that lines up with review positivity, which is why accounting for them matters. My next step is to build a regression model using genre, price, and age rating as predictors of `pct_pos_total`.

6. References

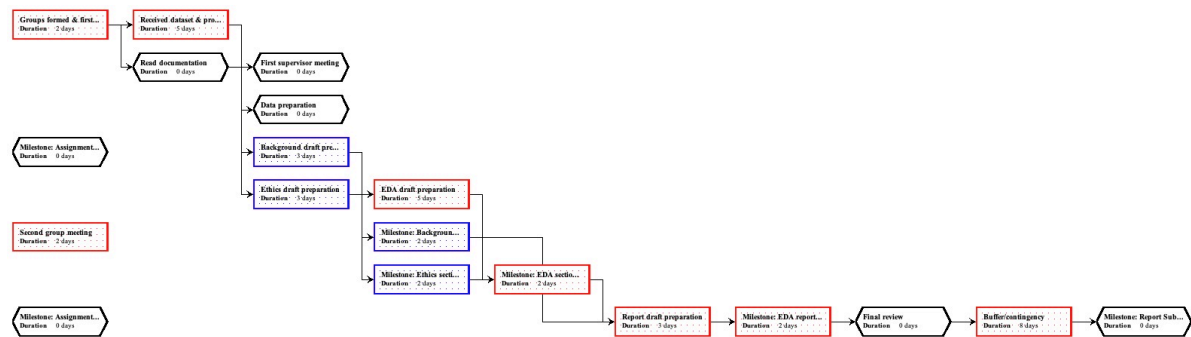
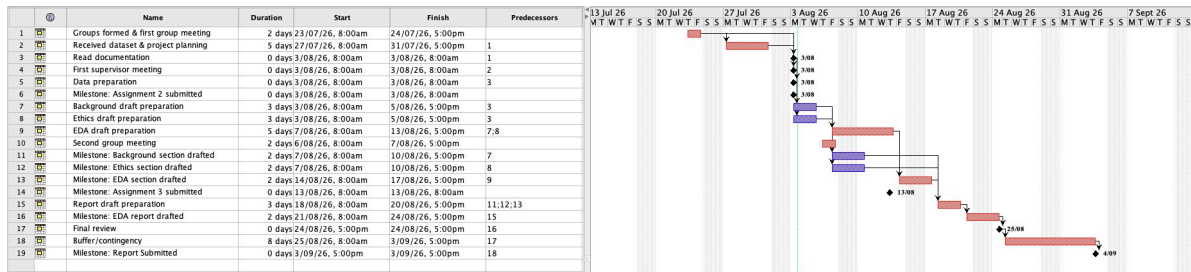
Ermilov, A. (2025). Steam games dataset.

- Kaggle.<https://www.kaggle.com/datasets/artermiloff/steam-games-dataset>

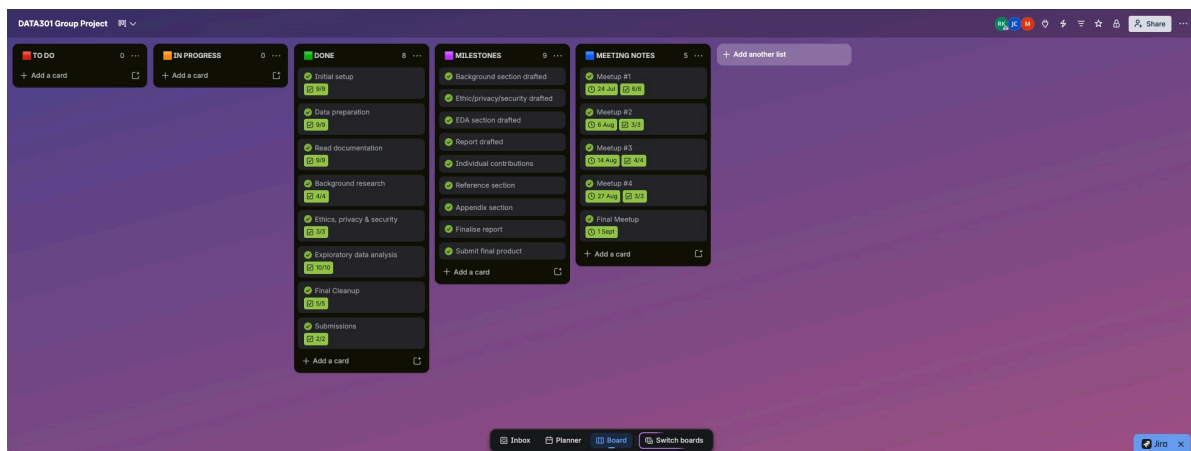
Kafadar, K. (2026). Boxplots: Definition, history, extensions, and examples.

- WIREs Computational Statistics, 18(3).<https://doi.org/10.1002/wics.70073>

7. Appendix A (Project Timeline)



8. Appendix B (Task Allocation)



9. Appendix C (Detailed Data Description)

Variable	Description	Type	Missing
appid	AppID of the game	int64	0
name	Name of the game	object	0
release_date	Game release date	object	0
required_age	Minimum age rating	int64	0
price	Base game price	float64	0
dlc_count	Number of DLC (extra downloadable content) per game	int64	0
detailed_description	Detailed game description	object	0
about_the_game	Game synopsis	object	0
short_description	Abbreviated game description	object	0
reviews	Submitted game reviews	object	0
header_image	Game icon image	object	0
website	Game website	object	0
support_url	Game support hyperlink	object	0
support_email	Game support email	object	0
windows	Boolean indicator for Windows support	bool	0
mac	Boolean indicator for Mac support	bool	0
linux	Boolean indicator for Linux support	bool	0
metacritic_score	Numeric Metacritic game score	int64	0
metacritic_url	Hyperlink to Metacritic review	object	0
achievements	Number of in-game achievements	int64	0
recommendations	Total positive user recommendations count	int64	0
notes	Additional game notes	object	0

Variable	Description	Type	Missing
supported_languages	List of available language options per game	object	0
full_audio_languages	List of language options for in-game audio	object	0
packages	Downloadable distribution files	object	0
developers	Game developers	object	0
publishers	Game publishers	object	0
categories	Game categories	object	0
genres	Game genres	object	0
screenshots	First available image on the game's Steam page	object	0
movies	Game promotional video or trailer	object	0
user_score	User score	int64	0
score_rank	Score rank	float64	89,579
positive	Number of positive user ratings	int64	0
negative	Number of negative user ratings	int64	0
estimated_owners	Estimated num of users owning game, given as a range	object	0
average_playtime_forever	Average total playtime in minutes for all users	int64	0
app id	AppID of the game	int64	0
name	Name of the game	object	0
release_date	Release date of the game	object	0
required_age	Minimum age rating	int64	0
average_playtime_2weeks	Average playtime in minutes for users over past 2 weeks	int64	0
median_playtime_forever	Median total playtime in minutes for all users	int64	0

Variable	Description	Type	Missing
median_play-time_2weeks	Median playtime in minutes for all users over past 2 weeks	int64	0
discount	Discount percent reduction from initial price	int64	0
peak_ccu	Highest number of concurrent users reached	int64	0
tags	List of specific game tags	object	0
pct_pos_total	Percentage of total positive reviews	int64	0
num_reviews_total	Count of total reviews	int64	0
pct_pos_recent	Percentage of total negative reviews	int64	0
num_reviews_recent	Number of relatively recently posted user reviews	int64	0